

MH1201 Linear Algebra II

Final Exam (Solutions)

AY 2024/2025, Semester 2

Quantitative Research Society @NTU

August 17, 2026

Overview

- Source paper: NTU AY2024/2025 Semester 2 MH1201 Linear Algebra II Final Examination.
- Solution style: the simplest direct computational solution is kept as Method 1; structural and theorem-heavy solutions are added as Method 2 onward.
- Each method is written as a standalone solution and is separated according to the original subparts, labelled (a), (b), etc.
- Page breaks are inserted before every solution, before Method 2 onward, and before every marking scheme.
- Main themes: nilpotent operators on polynomial spaces, diagonalisation by eigenbasis decomposition, matrix-induced inner products, Gram–Schmidt, Fourier coefficients, Frobenius isometries, and symmetric/skew-symmetric orthogonal decompositions.

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Question 1**(20 Marks)****Problem**

Let D denote the derivative linear transformation from $\mathcal{P}_2(\mathbb{R})$ to itself. Let I denote the identity linear transformation from $\mathcal{P}_2(\mathbb{R})$ to itself.

- (a) Compute the standard matrix representation of $D^2 + D + I$.

Note: Recall that the standard ordered basis for $\mathcal{P}_2(\mathbb{R})$ is $(1, X, X^2)$.

- (b) Using your response to Part (a), explain why $D^2 + D + I$ is invertible on $\mathcal{P}_2(\mathbb{R})$.

Note: Do not perform any hard computations for this part.

- (c) Using your response to Part (a), derive an explicit formula for $(D^2 + D + I)^{-1}$.

Note: Your formula must be expressed in terms of polynomials in its final form.

- (d) Therefore, determine all $P \in \mathcal{P}_2(\mathbb{R})$ such that

$$P'' + P' + P = -2 + 3X - 5X^2.$$

Solution

Method 1: Direct Standard-Matrix and Coefficient Computation

(a) Let the standard ordered basis be

$$\sigma = (1, X, X^2).$$

We compute the action of D , D^2 , and I on the basis vectors. Since

$$D(1) = 0, \quad D(X) = 1, \quad D(X^2) = 2X,$$

the matrix of D in the standard basis is

$$[D]_{\sigma}^{\sigma} = \begin{bmatrix} 0 & 1 & 0 \\ 0 & 0 & 2 \\ 0 & 0 & 0 \end{bmatrix}.$$

Also

$$D^2(1) = 0, \quad D^2(X) = 0, \quad D^2(X^2) = 2,$$

so

$$[D^2]_{\sigma}^{\sigma} = \begin{bmatrix} 0 & 0 & 2 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix}.$$

Therefore

$$[D^2 + D + I]_{\sigma}^{\sigma} = \begin{bmatrix} 1 & 1 & 2 \\ 0 & 1 & 2 \\ 0 & 0 & 1 \end{bmatrix}.$$

Hence

$$\boxed{[D^2 + D + I]_{\sigma}^{\sigma} = \begin{bmatrix} 1 & 1 & 2 \\ 0 & 1 & 2 \\ 0 & 0 & 1 \end{bmatrix}.$$

(b) The standard matrix representation obtained in Part (a) is

$$A := [D^2 + D + I]_{\sigma}^{\sigma} = \begin{bmatrix} 1 & 1 & 2 \\ 0 & 1 & 2 \\ 0 & 0 & 1 \end{bmatrix}.$$

This matrix is upper triangular with nonzero diagonal entries 1, 1, 1. Therefore

$$\det(A) = 1 \cdot 1 \cdot 1 = 1 \neq 0.$$

Equivalently, A has full rank:

$$\text{rank}(A) = 3 = \dim \mathcal{P}_2(\mathbb{R}).$$

Hence the standard matrix representation is invertible. Therefore the linear map $D^2 + D + I$ is invertible on $\mathcal{P}_2(\mathbb{R})$.

$$\boxed{D^2 + D + I \text{ is invertible.}}$$

(c) Let

$$Q(X) = a + bX + cX^2 \in \mathcal{P}_2(\mathbb{R}).$$

We seek

$$P(X) = \alpha + \beta X + \gamma X^2$$

such that

$$(D^2 + D + I)(P) = Q.$$

Since

$$P'(X) = \beta + 2\gamma X, \quad P''(X) = 2\gamma,$$

we get

$$(D^2 + D + I)(P) = (\alpha + \beta + 2\gamma) + (\beta + 2\gamma)X + \gamma X^2.$$

Equating this with $a + bX + cX^2$ gives

$$\gamma = c, \quad \beta + 2\gamma = b, \quad \alpha + \beta + 2\gamma = a.$$

Hence

$$\gamma = c, \quad \beta = b - 2c, \quad \alpha = a - b.$$

Therefore the inverse is

$$\boxed{(D^2 + D + I)^{-1}(a + bX + cX^2) = (a - b) + (b - 2c)X + cX^2.}$$

Equivalently, if $Q \in \mathcal{P}_2(\mathbb{R})$, then this is the same as

$$(D^2 + D + I)^{-1}(Q) = Q - Q'.$$

(d) We need to solve

$$(D^2 + D + I)(P) = -2 + 3X - 5X^2.$$

In the formula from Part (c), take

$$a = -2, \quad b = 3, \quad c = -5.$$

Then

$$(D^2 + D + I)^{-1}(-2 + 3X - 5X^2) = (-2 - 3) + (3 - 2(-5))X + (-5)X^2.$$

Hence

$$\boxed{P(X) = -5 + 13X - 5X^2.}$$

Method 2: One-Round Polynomial-Coefficient Map Computation

(a) Let

$$f(X) = a + bX + cX^2.$$

Then

$$f'(X) = b + 2cX, \quad f''(X) = 2c.$$

Hence

$$(D^2 + D + I)(f) = f'' + f' + f = (a + b + 2c) + (b + 2c)X + cX^2.$$

Therefore, relative to the standard ordered basis $\sigma = (1, X, X^2)$,

$$\begin{bmatrix} a \\ b \\ c \end{bmatrix} \mapsto \begin{bmatrix} a + b + 2c \\ b + 2c \\ c \end{bmatrix} = \begin{bmatrix} 1 & 1 & 2 \\ 0 & 1 & 2 \\ 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} a \\ b \\ c \end{bmatrix}.$$

Thus

$$[D^2 + D + I]_{\sigma}^{\sigma} = \begin{bmatrix} 1 & 1 & 2 \\ 0 & 1 & 2 \\ 0 & 0 & 1 \end{bmatrix}.$$

(b) The matrix

$$\begin{bmatrix} 1 & 1 & 2 \\ 0 & 1 & 2 \\ 0 & 0 & 1 \end{bmatrix}$$

is upper triangular with nonzero diagonal entries. Hence its determinant is the product of its diagonal entries:

$$1 \cdot 1 \cdot 1 = 1 \neq 0.$$

Therefore it has full rank and is invertible. Hence $D^2 + D + I$ is invertible.

$$D^2 + D + I \text{ is invertible.}$$

(c) Let the input polynomial be

$$q(X) = a + bX + cX^2.$$

From Part (a), if

$$(D^2 + D + I)(\alpha + \beta X + \gamma X^2) = a + bX + cX^2,$$

then

$$\gamma = c, \quad \beta = b - 2c, \quad \alpha = a - b.$$

Therefore

$$(D^2 + D + I)^{-1}(a + bX + cX^2) = (a - b) + (b - 2c)X + cX^2.$$

(d) Substituting $a = -2$, $b = 3$, and $c = -5$ into the inverse formula gives

$$P(X) = (-2 - 3) + (3 - 2(-5))X - 5X^2.$$

Therefore

$$P(X) = -5 + 13X - 5X^2.$$

Method 3: Nilpotent Geometric-Series Theorem

(a) In the standard basis $\sigma = (1, X, X^2)$, the derivative operator satisfies

$$[D]_{\sigma}^{\sigma} = \begin{bmatrix} 0 & 1 & 0 \\ 0 & 0 & 2 \\ 0 & 0 & 0 \end{bmatrix}.$$

Because $D^3 = 0$ on $\mathcal{P}_2(\mathbb{R})$, D is nilpotent of index at most 3. Directly,

$$[D^2]_{\sigma}^{\sigma} = \begin{bmatrix} 0 & 0 & 2 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix}.$$

Thus

$$[D^2 + D + I]_{\sigma}^{\sigma} = I + [D]_{\sigma}^{\sigma} + [D^2]_{\sigma}^{\sigma} = \boxed{\begin{bmatrix} 1 & 1 & 2 \\ 0 & 1 & 2 \\ 0 & 0 & 1 \end{bmatrix}}.$$

(b) Let

$$A = D^2 + D + I = I + D + D^2.$$

Since $D^3 = 0$, we have

$$(I + D + D^2)(I - D) = I - D + D - D^2 + D^2 - D^3 = I.$$

Similarly,

$$(I - D)(I + D + D^2) = I.$$

Therefore

$$A^{-1} = I - D.$$

Hence $A = D^2 + D + I$ is invertible.

$$\boxed{D^2 + D + I \text{ is invertible.}}$$

(c) Let

$$Q(X) = a + bX + cX^2.$$

From Part (b),

$$(D^2 + D + I)^{-1} = I - D.$$

Hence

$$(D^2 + D + I)^{-1}(Q) = Q - Q'.$$

Since

$$Q'(X) = b + 2cX,$$

we obtain

$$Q(X) - Q'(X) = (a + bX + cX^2) - (b + 2cX) = (a - b) + (b - 2c)X + cX^2.$$

Therefore

$$\boxed{(D^2 + D + I)^{-1}(a + bX + cX^2) = (a - b) + (b - 2c)X + cX^2.}$$

(d) Let

$$Q(X) = -2 + 3X - 5X^2.$$

Then

$$P = (I - D)Q = Q - Q'.$$

Since

$$Q'(X) = 3 - 10X,$$

we obtain

$$P(X) = -2 + 3X - 5X^2 - (3 - 10X) = -5 + 13X - 5X^2.$$

Hence

$$\boxed{P(X) = -5 + 13X - 5X^2.}$$

Method 4: Upper-Triangular Recursion Theorem

(a) For

$$P(X) = a + bX + cX^2,$$

we compute

$$P'(X) = b + 2cX, \quad P''(X) = 2c.$$

Therefore

$$P'' + P' + P = (a + b + 2c) + (b + 2c)X + cX^2.$$

Relative to $\sigma = (1, X, X^2)$, the coordinate vector $[P]_\sigma = (a, b, c)^\top$ is sent to

$$\begin{bmatrix} a + b + 2c \\ b + 2c \\ c \end{bmatrix} = \begin{bmatrix} 1 & 1 & 2 \\ 0 & 1 & 2 \\ 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} a \\ b \\ c \end{bmatrix}.$$

Hence

$$[D^2 + D + I]_\sigma^\sigma = \begin{bmatrix} 1 & 1 & 2 \\ 0 & 1 & 2 \\ 0 & 0 & 1 \end{bmatrix}.$$

(b) The coordinate formula is triangular:

$$(a, b, c) \mapsto (a + b + 2c, b + 2c, c).$$

The last output coordinate is exactly c , the second output coordinate determines b once c is known, and the first output coordinate determines a once b, c are known. Equivalently, the standard matrix is upper triangular with all diagonal entries nonzero, hence has full rank. Thus the map is invertible.

$$D^2 + D + I \text{ is invertible.}$$

(c) Given

$$Q(X) = a + bX + cX^2,$$

solve

$$(\alpha + \beta + 2\gamma) + (\beta + 2\gamma)X + \gamma X^2 = a + bX + cX^2.$$

Equating coefficients from highest degree to lowest degree gives

$$\gamma = c, \quad \beta = b - 2c, \quad \alpha = a - b.$$

Therefore

$$(D^2 + D + I)^{-1}(a + bX + cX^2) = (a - b) + (b - 2c)X + cX^2.$$

(d) For

$$-2 + 3X - 5X^2,$$

we have $a = -2$, $b = 3$, and $c = -5$. Hence

$$\gamma = -5, \quad \beta = 3 - 2(-5) = 13, \quad \alpha = -2 - 3 = -5.$$

Therefore

$$P(X) = -5 + 13X - 5X^2.$$

Marking Scheme

- Part (a), 5 marks: correct matrices for D and D^2 , and correct final matrix for $D^2 + D + I$.
- Part (b), 4 marks: identifies upper-triangular form or nonzero determinant, and concludes invertibility.
- Part (c), 6 marks: derives the inverse formula either by coefficient comparison or by $D^3 = 0$; final formula must be polynomial-valued.
- Part (d), 5 marks: applies the inverse to the given polynomial and obtains $-5 + 13X - 5X^2$.

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Question 2**(30 Marks)****Problem**

Define a linear transformation $T \in \mathcal{L}(\mathcal{P}_2(\mathbb{R}))$ as follows:

$$\forall P \in \mathcal{P}_2(\mathbb{R}) : \quad T(P) := P(0) + P(1)(X + X^2).$$

- (a) Compute the standard matrix representation of T .

Note: Recall that the standard ordered basis for $\mathcal{P}_2(\mathbb{R})$ is $(1, X, X^2)$.

- (b) Using your response to Part (a), compute all the eigenvalues of T .
- (c) Using your response to Part (b), determine all the eigenspaces of T .
- (d) Using your response to Part (b) or Part (c), explain why T is diagonalizable.
- (e) Therefore, compute $T^{10}(1 + 2X + 4X^2)$.

Solution

Method 1: Direct Matrix, Eigenvalue, and Eigenvector Computation

(a) Let

$$P(X) = a + bX + cX^2.$$

Then

$$P(0) = a, \quad P(1) = a + b + c.$$

Therefore

$$T(P) = P(0) + P(1)(X + X^2) = a + (a + b + c)X + (a + b + c)X^2.$$

Thus, relative to the standard ordered basis $\sigma = (1, X, X^2)$,

$$\begin{bmatrix} a \\ b \\ c \end{bmatrix} \mapsto \begin{bmatrix} a \\ a + b + c \\ a + b + c \end{bmatrix} = \begin{bmatrix} 1 & 0 & 0 \\ 1 & 1 & 1 \\ 1 & 1 & 1 \end{bmatrix} \begin{bmatrix} a \\ b \\ c \end{bmatrix}.$$

Hence

$$[T]_{\sigma}^{\sigma} = \begin{bmatrix} 1 & 0 & 0 \\ 1 & 1 & 1 \\ 1 & 1 & 1 \end{bmatrix}.$$

(b) Let

$$A = [T]_{\sigma}^{\sigma} = \begin{bmatrix} 1 & 0 & 0 \\ 1 & 1 & 1 \\ 1 & 1 & 1 \end{bmatrix}.$$

Then

$$A - \lambda I = \begin{bmatrix} 1 - \lambda & 0 & 0 \\ 1 & 1 - \lambda & 1 \\ 1 & 1 & 1 - \lambda \end{bmatrix}.$$

Expanding along the first row gives

$$\det(A - \lambda I) = (1 - \lambda) \det \begin{bmatrix} 1 - \lambda & 1 \\ 1 & 1 - \lambda \end{bmatrix}.$$

Hence

$$\det(A - \lambda I) = (1 - \lambda)((1 - \lambda)^2 - 1) = (1 - \lambda)\lambda(\lambda - 2).$$

Therefore the eigenvalues are

$$\boxed{0, 1, 2}.$$

(c) We solve for each eigenvalue.

For $\lambda = 0$, solve $A(a, b, c)^T = 0$:

$$a = 0, \quad a + b + c = 0.$$

Therefore $a = 0$ and $c = -b$, so

$$E_0 = \text{span}\{X - X^2\}.$$

For $\lambda = 1$, solve $(A - I)(a, b, c)^T = 0$. This gives

$$a + c = 0, \quad a + b = 0.$$

Hence $b = -a$ and $c = -a$, so

$$E_1 = \text{span}\{1 - X - X^2\}.$$

For $\lambda = 2$, solve $(A - 2I)(a, b, c)^T = 0$. This gives

$$a = 0, \quad b = c.$$

Hence

$$E_2 = \text{span}\{X + X^2\}.$$

Therefore

$$\boxed{E_0 = \text{span}\{X - X^2\}, \quad E_1 = \text{span}\{1 - X - X^2\}, \quad E_2 = \text{span}\{X + X^2\}.$$

- (d) Since T has three distinct eigenvalues on the 3-dimensional vector space $\mathcal{P}_2(\mathbb{R})$, it has three linearly independent eigenvectors. Therefore T is diagonalizable.

$$\boxed{T \text{ is diagonalizable.}}$$

- (e) Decompose

$$1 + 2X + 4X^2 = \alpha(1 - X - X^2) + \beta(X - X^2) + \gamma(X + X^2).$$

Comparing coefficients gives

$$\alpha = 1, \quad -\alpha + \beta + \gamma = 2, \quad -\alpha - \beta + \gamma = 4.$$

Thus

$$\beta + \gamma = 3, \quad -\beta + \gamma = 5,$$

so

$$\beta = -1, \quad \gamma = 4.$$

Hence

$$1 + 2X + 4X^2 = (1 - X - X^2) - (X - X^2) + 4(X + X^2).$$

Applying T^{10} gives

$$T^{10}(1 - X - X^2) = 1^{10}(1 - X - X^2) = 1 - X - X^2,$$

$$T^{10}(X - X^2) = 0^{10}(X - X^2) = 0,$$

and

$$T^{10}(X + X^2) = 2^{10}(X + X^2) = 1024(X + X^2).$$

Therefore

$$T^{10}(1 + 2X + 4X^2) = 1 - X - X^2 + 4 \cdot 1024(X + X^2).$$

Hence

$$\boxed{T^{10}(1 + 2X + 4X^2) = 1 + 4095X + 4095X^2.}$$

Method 2: Range-Kernel Decomposition Theorem

(a) Let

$$v := X + X^2.$$

Then

$$T(P) = P(0) + P(1)v.$$

Applying this to $1, X, X^2$ gives

$$T(1) = 1 + v, \quad T(X) = v, \quad T(X^2) = v.$$

Therefore

$$[T]_{\sigma}^{\sigma} = \begin{bmatrix} 1 & 0 & 0 \\ 1 & 1 & 1 \\ 1 & 1 & 1 \end{bmatrix}.$$

(b) The range of T is contained in

$$U := \text{span}\{1, v\}.$$

On U , with ordered basis $(1, v)$, we have

$$T(1) = 1 + v, \quad T(v) = T(X + X^2) = 2v.$$

Thus

$$[T|_U]_{(1,v)}^{(1,v)} = \begin{bmatrix} 1 & 0 \\ 1 & 2 \end{bmatrix}.$$

Hence $T|_U$ has eigenvalues 1 and 2. Also

$$\text{null}(T) = \{P \in \mathcal{P}_2(\mathbb{R}) : P(0) = 0, \text{ and } P(1) = 0\}.$$

A nonzero polynomial in this nullspace is

$$X - X^2,$$

so 0 is also an eigenvalue. Therefore the eigenvalues of T are

$$\boxed{0, 1, 2}.$$

(c) The 0-eigenspace is the nullspace:

$$E_0 = \text{null}(T) = \text{span}\{X - X^2\}.$$

For the eigenvalue 1, solve inside $U = \text{span}\{1, v\}$:

$$T(\alpha + \gamma v) = \alpha + (\alpha + 2\gamma)v.$$

If this equals $\alpha + \gamma v$, then

$$\alpha + 2\gamma = \gamma,$$

so $\gamma = -\alpha$. Hence

$$E_1 = \text{span}\{1 - v\} = \text{span}\{1 - X - X^2\}.$$

For the eigenvalue 2, we need

$$\alpha + (\alpha + 2\gamma)v = 2\alpha + 2\gamma v.$$

Hence $\alpha = 0$, and so

$$E_2 = \text{span}\{v\} = \text{span}\{X + X^2\}.$$

Therefore

$$\boxed{E_0 = \text{span}\{X - X^2\}, \quad E_1 = \text{span}\{1 - X - X^2\}, \quad E_2 = \text{span}\{X + X^2\}.$$

- (d) The three eigenspaces contain nonzero eigenvectors belonging to three distinct eigenvalues. Therefore those eigenvectors are linearly independent. Since $\dim \mathcal{P}_2(\mathbb{R}) = 3$, they form a basis. Hence T is diagonalizable.

T is diagonalizable.

- (e) Using the eigenbasis found in Part (c), write

$$1 + 2X + 4X^2 = (1 - X - X^2) - (X - X^2) + 4(X + X^2).$$

Therefore

$$T^{10}(1 + 2X + 4X^2) = 1^{10}(1 - X - X^2) - 0^{10}(X - X^2) + 4 \cdot 2^{10}(X + X^2).$$

Since $10 > 0$, the 0^{10} term is 0. Hence

$$T^{10}(1 + 2X + 4X^2) = 1 + 4095X + 4095X^2.$$

Method 3: Spectral Projector Theorem

(a) As before, from

$$T(1) = 1 + X + X^2, \quad T(X) = X + X^2, \quad T(X^2) = X + X^2,$$

we obtain

$$[T]_{\sigma}^{\sigma} = \begin{bmatrix} 1 & 0 & 0 \\ 1 & 1 & 1 \\ 1 & 1 & 1 \end{bmatrix}.$$

(b) The characteristic polynomial is

$$\det([T] - \lambda I) = (1 - \lambda)\lambda(\lambda - 2).$$

Hence the eigenvalues are

$$\boxed{0, 1, 2}.$$

(c) Since the eigenvalues are distinct, the spectral projectors are

$$\Pi_0 = \frac{(T - I)(T - 2I)}{(0 - 1)(0 - 2)} = \frac{1}{2}(T - I)(T - 2I),$$

$$\Pi_1 = \frac{T(T - 2I)}{(1 - 0)(1 - 2)} = -T(T - 2I),$$

$$\Pi_2 = \frac{T(T - I)}{(2 - 0)(2 - 1)} = \frac{1}{2}T(T - I).$$

The ranges of these projectors are the corresponding eigenspaces. Applying the direct computations from the projectors gives

$$\text{range}(\Pi_0) = \text{span}\{X - X^2\},$$

$$\text{range}(\Pi_1) = \text{span}\{1 - X - X^2\},$$

$$\text{range}(\Pi_2) = \text{span}\{X + X^2\}.$$

Hence

$$\boxed{E_0 = \text{span}\{X - X^2\}, \quad E_1 = \text{span}\{1 - X - X^2\}, \quad E_2 = \text{span}\{X + X^2\}.$$

(d) Because the characteristic polynomial splits into three distinct linear factors, the minimal polynomial also has no repeated root. Therefore T is diagonalizable.

$$\boxed{T \text{ is diagonalizable.}}$$

(e) Let

$$P = 1 + 2X + 4X^2.$$

The spectral decomposition is

$$P = \Pi_0 P + \Pi_1 P + \Pi_2 P.$$

We compute

$$\Pi_1 P = 1 - X - X^2, \quad \Pi_2 P = 4(X + X^2), \quad \Pi_0 P = -(X - X^2).$$

Therefore

$$T^{10}P = 0^{10}\Pi_0P + 1^{10}\Pi_1P + 2^{10}\Pi_2P.$$

Since $10 > 0$, this becomes

$$T^{10}P = 1 - X - X^2 + 1024 \cdot 4(X + X^2).$$

Hence

$$\boxed{T^{10}(1 + 2X + 4X^2) = 1 + 4095X + 4095X^2.}$$

Marking Scheme

- Part (a), 5 marks: correctly evaluates T on $1, X, X^2$ and forms the matrix.
- Part (b), 6 marks: computes the characteristic polynomial and identifies eigenvalues $0, 1, 2$.
- Part (c), 9 marks: solves each eigenspace accurately.
- Part (d), 4 marks: explains diagonalizability using distinct eigenvalues or an eigenbasis.
- Part (e), 6 marks: decomposes the polynomial into eigenvectors and applies T^{10} correctly.

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Question 3**(30 Marks)****Problem**

Define a function $\langle \cdot, \cdot \rangle : \mathbb{R}^{3,1} \times \mathbb{R}^{3,1} \rightarrow \mathbb{R}$ as follows:

$$\forall x, y \in \mathbb{R}^{3,1} : \quad \langle x, y \rangle := x^T \begin{bmatrix} 1 & 0 & 1 \\ 0 & 2 & 0 \\ 1 & 0 & 2 \end{bmatrix} y.$$

Let σ denote the standard ordered basis for $\mathbb{R}^{3,1}$.

- (a) Verify that $\langle \cdot, \cdot \rangle$ satisfies the axioms for an inner product on $\mathbb{R}^{3,1}$.
- (b) Apply Gram–Schmidt to σ to obtain an orthonormal basis β for $(\mathbb{R}^{3,1}, \langle \cdot, \cdot \rangle)$.
- (c) Determine the β -Fourier coefficients of

$$\begin{bmatrix} 1 \\ 2 \\ 3 \end{bmatrix}$$

with respect to $(\mathbb{R}^{3,1}, \langle \cdot, \cdot \rangle)$.

Solution

Let

$$H := \begin{bmatrix} 1 & 0 & 1 \\ 0 & 2 & 0 \\ 1 & 0 & 2 \end{bmatrix}.$$

Thus

$$\langle x, y \rangle = x^\top H y.$$

Method 1: Direct Axiom Check and Gram–Schmidt Computation

(a) We verify the inner product axioms.

Linearity in each variable. For $a, b \in \mathbb{R}$ and $x_1, x_2, y \in \mathbb{R}^{3,1}$,

$$\langle ax_1 + bx_2, y \rangle = (ax_1 + bx_2)^\top H y = ax_1^\top H y + bx_2^\top H y = a \langle x_1, y \rangle + b \langle x_2, y \rangle.$$

Similarly,

$$\langle x, ay_1 + by_2 \rangle = a \langle x, y_1 \rangle + b \langle x, y_2 \rangle.$$

Symmetry. Since $H^\top = H$, we have

$$\langle x, y \rangle = x^\top H y = (x^\top H y)^\top = y^\top H^\top x = y^\top H x = \langle y, x \rangle.$$

Positive definiteness. For $x = (x_1, x_2, x_3)^\top$,

$$\langle x, x \rangle = x^\top H x = x_1^2 + 2x_1x_3 + 2x_2^2 + 2x_3^2.$$

Completing the square gives

$$\langle x, x \rangle = (x_1 + x_3)^2 + 2x_2^2 + x_3^2.$$

This is nonnegative. If it equals 0, then

$$x_1 + x_3 = 0, \quad x_2 = 0, \quad x_3 = 0,$$

which implies $x_1 = x_2 = x_3 = 0$. Hence

$$\langle x, x \rangle > 0 \quad \text{for all } x \neq 0.$$

Therefore

$$\boxed{\langle \cdot, \cdot \rangle \text{ is an inner product on } \mathbb{R}^{3,1}.}$$

(b) Let

$$e_1 = \begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix}, \quad e_2 = \begin{bmatrix} 0 \\ 1 \\ 0 \end{bmatrix}, \quad e_3 = \begin{bmatrix} 0 \\ 0 \\ 1 \end{bmatrix}.$$

Since $\langle e_i, e_j \rangle = H_{ij}$, we have

$$\langle e_1, e_1 \rangle = 1, \quad \langle e_1, e_2 \rangle = 0, \quad \langle e_1, e_3 \rangle = 1,$$

$$\langle e_2, e_2 \rangle = 2, \quad \langle e_2, e_3 \rangle = 0, \quad \langle e_3, e_3 \rangle = 2.$$

Gram–Schmidt gives

$$u_1 = e_1.$$

Since $\langle e_1, e_2 \rangle = 0$,

$$u_2 = e_2.$$

For the third vector,

$$u_3 = e_3 - \frac{\langle u_1, e_3 \rangle}{\langle u_1, u_1 \rangle} u_1 - \frac{\langle u_2, e_3 \rangle}{\langle u_2, u_2 \rangle} u_2 = e_3 - e_1.$$

Thus

$$u_3 = \begin{bmatrix} -1 \\ 0 \\ 1 \end{bmatrix}.$$

The norms are

$$\|u_1\| = 1, \quad \|u_2\| = \sqrt{2}, \quad \|u_3\| = 1.$$

Hence an orthonormal basis is

$$\beta = \left(\begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix}, \frac{1}{\sqrt{2}} \begin{bmatrix} 0 \\ 1 \\ 0 \end{bmatrix}, \begin{bmatrix} -1 \\ 0 \\ 1 \end{bmatrix} \right).$$

(c) Let

$$v = \begin{bmatrix} 1 \\ 2 \\ 3 \end{bmatrix}.$$

Since $\beta = (\beta_1, \beta_2, \beta_3)$ is orthonormal, the Fourier coefficients are

$$c_i = \langle \beta_i, v \rangle.$$

First compute

$$Hv = \begin{bmatrix} 1 & 0 & 1 \\ 0 & 2 & 0 \\ 1 & 0 & 2 \end{bmatrix} \begin{bmatrix} 1 \\ 2 \\ 3 \end{bmatrix} = \begin{bmatrix} 4 \\ 4 \\ 7 \end{bmatrix}.$$

Therefore

$$c_1 = e_1^\top Hv = 4,$$

$$c_2 = \left(\frac{1}{\sqrt{2}} e_2 \right)^\top Hv = \frac{4}{\sqrt{2}} = 2\sqrt{2},$$

and

$$c_3 = (e_3 - e_1)^\top Hv = 7 - 4 = 3.$$

Hence

$$(c_1, c_2, c_3) = (4, 2\sqrt{2}, 3).$$

Method 2: Sylvester Criterion and Gram-Matrix Theorem

- (a) The theorem used is: if H is symmetric positive definite, then $\langle x, y \rangle = x^T H y$ defines an inner product on $\mathbb{R}^n$.

The matrix H is symmetric because

$$H^T = H.$$

Its leading principal minors are

$$\begin{aligned}\Delta_1 &= 1 > 0, \\ \Delta_2 &= \det \begin{bmatrix} 1 & 0 \\ 0 & 2 \end{bmatrix} = 2 > 0,\end{aligned}$$

and

$$\Delta_3 = \det \begin{bmatrix} 1 & 0 & 1 \\ 0 & 2 & 0 \\ 1 & 0 & 2 \end{bmatrix} = 2 \det \begin{bmatrix} 1 & 1 \\ 1 & 2 \end{bmatrix} = 2 > 0.$$

By Sylvester's criterion, H is positive definite. Therefore $x^T H y$ defines an inner product.

$$\boxed{\langle \cdot, \cdot \rangle \text{ is an inner product.}}$$

- (b) The Gram matrix of the standard basis under this inner product is H itself, because

$$\langle e_i, e_j \rangle = H_{ij}.$$

Therefore

$$\begin{aligned}\langle e_1, e_1 \rangle &= 1, & \langle e_1, e_2 \rangle &= 0, & \langle e_1, e_3 \rangle &= 1, \\ \langle e_2, e_2 \rangle &= 2, & \langle e_2, e_3 \rangle &= 0, & \langle e_3, e_3 \rangle &= 2.\end{aligned}$$

Hence Gram-Schmidt gives

$$u_1 = e_1, \quad u_2 = e_2, \quad u_3 = e_3 - e_1.$$

Normalising gives

$$\boxed{\beta = \left(e_1, \frac{e_2}{\sqrt{2}}, e_3 - e_1 \right)}.$$

- (c) For

$$v = \begin{bmatrix} 1 \\ 2 \\ 3 \end{bmatrix},$$

the Gram-matrix method gives

$$Hv = \begin{bmatrix} 4 \\ 4 \\ 7 \end{bmatrix}.$$

Hence

$$\langle e_1, v \rangle = 4, \quad \langle e_2/\sqrt{2}, v \rangle = 2\sqrt{2}, \quad \langle e_3 - e_1, v \rangle = 3.$$

Therefore

$$\boxed{(c_1, c_2, c_3) = (4, 2\sqrt{2}, 3)}.$$

Method 3: Cholesky Whitening Theorem

(a) Observe that

$$H = R^T R, \quad R = \begin{bmatrix} 1 & 0 & 1 \\ 0 & \sqrt{2} & 0 \\ 0 & 0 & 1 \end{bmatrix}.$$

Therefore

$$\langle x, y \rangle = x^T H y = x^T R^T R y = (R x) \cdot (R y).$$

Thus the given inner product is the ordinary Euclidean dot product after the invertible change of variables

$$x \mapsto R x.$$

Since R is invertible, $R x = 0$ implies $x = 0$. Hence

$$\langle x, x \rangle = (R x) \cdot (R x) = \|R x\|_2^2 \geq 0,$$

with equality if and only if $x = 0$. Bilinearity and symmetry follow from bilinearity and symmetry of the Euclidean dot product. Therefore

$\langle \cdot, \cdot \rangle$ is an inner product.

(b) The whitening identity

$$\langle x, y \rangle = (R x) \cdot (R y)$$

means that applying Gram–Schmidt to $\{e_1, e_2, e_3\}$ under $\langle \cdot, \cdot \rangle$ is equivalent to applying ordinary Euclidean Gram–Schmidt to

$$R e_1, \quad R e_2, \quad R e_3,$$

then pulling the result back by R^{-1} .

First compute

$$R e_1 = \begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix}, \quad R e_2 = \begin{bmatrix} 0 \\ \sqrt{2} \\ 0 \end{bmatrix}, \quad R e_3 = \begin{bmatrix} 1 \\ 0 \\ 1 \end{bmatrix}.$$

Let

$$w_1 = R e_1, \quad w_2 = R e_2, \quad w_3 = R e_3.$$

We now apply ordinary Euclidean Gram–Schmidt to (w_1, w_2, w_3) .

For the first vector,

$$\tilde{q}_1 = w_1 = \begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix}, \quad \|\tilde{q}_1\|_2 = 1,$$

so

$$q_1 = \frac{\tilde{q}_1}{\|\tilde{q}_1\|_2} = \begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix}.$$

For the second vector,

$$\tilde{q}_2 = w_2 - (w_2 \cdot q_1) q_1.$$

Since

$$w_2 \cdot q_1 = \begin{bmatrix} 0 \\ \sqrt{2} \\ 0 \end{bmatrix} \cdot \begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix} = 0,$$

we get

$$\tilde{q}_2 = w_2 = \begin{bmatrix} 0 \\ \sqrt{2} \\ 0 \end{bmatrix}.$$

Hence

$$\|\tilde{q}_2\|_2 = \sqrt{2}, \quad q_2 = \frac{\tilde{q}_2}{\|\tilde{q}_2\|_2} = \begin{bmatrix} 0 \\ 1 \\ 0 \end{bmatrix}.$$

For the third vector,

$$\tilde{q}_3 = w_3 - (w_3 \cdot q_1)q_1 - (w_3 \cdot q_2)q_2.$$

Now

$$w_3 \cdot q_1 = \begin{bmatrix} 1 \\ 0 \\ 1 \end{bmatrix} \cdot \begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix} = 1,$$

and

$$w_3 \cdot q_2 = \begin{bmatrix} 1 \\ 0 \\ 1 \end{bmatrix} \cdot \begin{bmatrix} 0 \\ 1 \\ 0 \end{bmatrix} = 0.$$

Therefore

$$\tilde{q}_3 = \begin{bmatrix} 1 \\ 0 \\ 1 \end{bmatrix} - 1 \begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix} - 0 \begin{bmatrix} 0 \\ 1 \\ 0 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 1 \end{bmatrix}.$$

Since

$$\|\tilde{q}_3\|_2 = 1,$$

we have

$$q_3 = \frac{\tilde{q}_3}{\|\tilde{q}_3\|_2} = \begin{bmatrix} 0 \\ 0 \\ 1 \end{bmatrix}.$$

Thus ordinary Euclidean Gram–Schmidt gives

$$(q_1, q_2, q_3) = \left(\begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix}, \begin{bmatrix} 0 \\ 1 \\ 0 \end{bmatrix}, \begin{bmatrix} 0 \\ 0 \\ 1 \end{bmatrix} \right).$$

Now pull this Euclidean orthonormal basis back to the original space. Since

$$R^{-1} = \begin{bmatrix} 1 & 0 & -1 \\ 0 & \frac{1}{\sqrt{2}} & 0 \\ 0 & 0 & 1 \end{bmatrix},$$

we get

$$R^{-1}q_1 = \begin{bmatrix} 1 & 0 & -1 \\ 0 & \frac{1}{\sqrt{2}} & 0 \\ 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix} = e_1,$$

$$R^{-1}q_2 = \begin{bmatrix} 1 & 0 & -1 \\ 0 & \frac{1}{\sqrt{2}} & 0 \\ 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} 0 \\ 1 \\ 0 \end{bmatrix} = \frac{e_2}{\sqrt{2}},$$

and

$$R^{-1}q_3 = \begin{bmatrix} 1 & 0 & -1 \\ 0 & \frac{1}{\sqrt{2}} & 0 \\ 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} 0 \\ 0 \\ 1 \end{bmatrix} = -e_1 + e_3.$$

Hence

$$\beta = \left(e_1, \frac{e_2}{\sqrt{2}}, e_3 - e_1 \right).$$

Finally, this basis is orthonormal with respect to $\langle \cdot, \cdot \rangle$ because

$$R\beta_i = q_i \quad (i = 1, 2, 3),$$

and hence

$$\langle \beta_i, \beta_j \rangle = (R\beta_i) \cdot (R\beta_j) = q_i \cdot q_j = \delta_{ij}.$$

(c) Let

$$v = \begin{bmatrix} 1 \\ 2 \\ 3 \end{bmatrix}.$$

Since

$$R\beta_1 = q_1, \quad R\beta_2 = q_2, \quad R\beta_3 = q_3,$$

the β -Fourier coefficients of v under $\langle \cdot, \cdot \rangle$ equal the ordinary Euclidean Fourier coefficients of Rv under the orthonormal basis (q_1, q_2, q_3) .

Compute

$$Rv = \begin{bmatrix} 1 & 0 & 1 \\ 0 & \sqrt{2} & 0 \\ 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} 1 \\ 2 \\ 3 \end{bmatrix} = \begin{bmatrix} 4 \\ 2\sqrt{2} \\ 3 \end{bmatrix}.$$

Since (q_1, q_2, q_3) is the standard Euclidean orthonormal basis, the coefficients are directly the entries of Rv :

$$c_1 = 4, \quad c_2 = 2\sqrt{2}, \quad c_3 = 3.$$

Equivalently,

$$c_i = \langle \beta_i, v \rangle = (R\beta_i) \cdot (Rv) = q_i \cdot (Rv).$$

Thus

$$(c_1, c_2, c_3) = (4, 2\sqrt{2}, 3).$$

Marking Scheme

- Part (a), 10 marks: verifies bilinearity, symmetry, and positive definiteness; positive definiteness may be shown by completing the square or Sylvester's criterion.
- Part (b), 12 marks: applies Gram–Schmidt correctly and normalises the basis vectors.
- Part (c), 8 marks: computes Fourier coefficients using $c_i = \langle \beta_i, v \rangle$ and gives $(4, 2\sqrt{2}, 3)$.

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Question 4**(20 Marks)****Problem**

Let n be a natural number. Recall that the Frobenius inner product $\langle \cdot, \cdot \rangle_F$ on $\mathbb{R}^{n,n}$ is defined as follows:

$$\forall A, B \in \mathbb{R}^{n,n} : \quad \langle A, B \rangle_F := \text{trace}(A^T B).$$

- (a) Show that if $M \in \mathbb{R}^{n,n}$ satisfies the property that $M^T M = I_n$, then

$$\forall A, B \in \mathbb{R}^{n,n} : \quad \langle MA, MB \rangle_F = \langle A, B \rangle_F.$$

- (b) Define vector subspaces V and W of $\mathbb{R}^{n,n}$ as follows:

$$V := \{A \in \mathbb{R}^{n,n} : A^T = A\}, \quad W := \{B \in \mathbb{R}^{n,n} : B^T = -B\}.$$

Show that $W = V^\perp$ with respect to $(\mathbb{R}^{n,n}, \langle \cdot, \cdot \rangle_F)$.

Solution**Method 1: Direct Trace Computation and Dimension Counting**

- (a) Let $A, B \in \mathbb{R}^{n,n}$. Then

$$\langle MA, MB \rangle_F = \text{trace}((MA)^T (MB)).$$

Since

$$(MA)^T = A^T M^T,$$

we get

$$\langle MA, MB \rangle_F = \text{trace}(A^T M^T MB).$$

Because $M^T M = I_n$,

$$\langle MA, MB \rangle_F = \text{trace}(A^T B) = \langle A, B \rangle_F.$$

Therefore

$$\boxed{\langle MA, MB \rangle_F = \langle A, B \rangle_F.}$$

- (b) First show that

$$W \subseteq V^\perp.$$

Let $S \in V$ and $K \in W$. Thus

$$S^T = S, \quad K^T = -K.$$

Then

$$\langle S, K \rangle_F = \text{trace}(S^T K) = \text{trace}(SK).$$

Since trace is invariant under transpose,

$$\text{trace}(SK) = \text{trace}((SK)^T) = \text{trace}(K^T S^T) = \text{trace}((-K)S) = -\text{trace}(KS).$$

Using $\text{trace}(KS) = \text{trace}(SK)$, we obtain

$$\text{trace}(SK) = -\text{trace}(SK),$$

and hence

$$\text{trace}(SK) = 0.$$

Therefore

$$\langle S, K \rangle_F = 0,$$

so $K \in V^\perp$. Hence

$$W \subseteq V^\perp.$$

Next, compute dimensions. A symmetric matrix has n free diagonal entries and $n(n-1)/2$ free off-diagonal pairs, so

$$\dim V = \frac{n(n+1)}{2}.$$

Since $\dim \mathbb{R}^{n,n} = n^2$,

$$\dim V^\perp = n^2 - \dim V = n^2 - \frac{n(n+1)}{2} = \frac{n(n-1)}{2}.$$

A skew-symmetric matrix has zero diagonal and $n(n-1)/2$ free off-diagonal pairs, so

$$\dim W = \frac{n(n-1)}{2}.$$

Thus

$$W \subseteq V^\perp \quad \text{and} \quad \dim W = \dim V^\perp.$$

Therefore

$$\boxed{W = V^\perp.}$$

Method 2: Direct Orthogonal-Complement Characterisation

(a) Let $A, B \in \mathbb{R}^{n,n}$. Then

$$\langle MA, MB \rangle_F = \text{trace}((MA)^\top(MB)) = \text{trace}(A^\top M^\top MB).$$

Since $M^\top M = I_n$, this becomes

$$\langle MA, MB \rangle_F = \text{trace}(A^\top B) = \langle A, B \rangle_F.$$

Hence

$$\boxed{\langle MA, MB \rangle_F = \langle A, B \rangle_F.}$$

(b) We prove $V^\perp = W$ directly from the definition of orthogonal complement.

First, let $K \in W$, so $K^\top = -K$. For any $S \in V$, so $S^\top = S$, we have

$$\langle S, K \rangle_F = \text{trace}(S^\top K) = \text{trace}(SK).$$

Using invariance of trace under transpose and cyclicity of trace,

$$\text{trace}(SK) = \text{trace}((SK)^\top) = \text{trace}(K^\top S^\top) = \text{trace}((-K)S) = -\text{trace}(KS) = -\text{trace}(SK).$$

Therefore $\text{trace}(SK) = 0$, so

$$\langle S, K \rangle_F = 0 \quad \text{for all } S \in V.$$

Hence

$$W \subseteq V^\perp.$$

Conversely, let $C = (c_{ij}) \in V^\perp$. We show that C must be skew-symmetric. Since every diagonal matrix unit E_{ii} is symmetric and lies in V ,

$$0 = \langle E_{ii}, C \rangle_F = \text{trace}(E_{ii}^\top C) = c_{ii}.$$

Hence

$$c_{ii} = 0 \quad (1 \leq i \leq n).$$

Next, for $i < j$, the matrix

$$E_{ij} + E_{ji}$$

is symmetric and therefore belongs to V . Since $C \in V^\perp$,

$$0 = \langle E_{ij} + E_{ji}, C \rangle_F = c_{ij} + c_{ji}.$$

Hence

$$c_{ji} = -c_{ij} \quad (i < j).$$

Combining the diagonal condition $c_{ii} = 0$ with the off-diagonal condition $c_{ji} = -c_{ij}$ gives

$$C^\top = -C.$$

Therefore $C \in W$, so

$$V^\perp \subseteq W.$$

Since both inclusions hold,

$$\boxed{W = V^\perp.}$$

Method 3: Orthogonal Direct-Sum Decomposition Theorem

- (a) The Frobenius inner product can be interpreted columnwise. Let a_j and b_j be the j -th columns of A and B . Then

$$\langle A, B \rangle_F = \sum_{j=1}^n a_j \cdot b_j.$$

Since $M^T M = I_n$, the map $x \mapsto Mx$ preserves the Euclidean dot product:

$$(Mx) \cdot (My) = x^T M^T M y = x \cdot y.$$

Therefore

$$\langle MA, MB \rangle_F = \sum_{j=1}^n (Ma_j) \cdot (Mb_j) = \sum_{j=1}^n a_j \cdot b_j = \langle A, B \rangle_F.$$

Hence

$$\boxed{\langle MA, MB \rangle_F = \langle A, B \rangle_F.}$$

- (b) Every matrix $A \in \mathbb{R}^{n,n}$ decomposes uniquely as

$$A = \frac{A + A^T}{2} + \frac{A - A^T}{2}.$$

The first term is symmetric because

$$\left(\frac{A + A^T}{2} \right)^T = \frac{A^T + A}{2},$$

and the second term is skew-symmetric because

$$\left(\frac{A - A^T}{2} \right)^T = \frac{A^T - A}{2} = -\frac{A - A^T}{2}.$$

Thus

$$\mathbb{R}^{n,n} = V + W.$$

Moreover, if $S \in V$ and $K \in W$, then as shown in Method 1,

$$\langle S, K \rangle_F = 0.$$

Therefore

$$\mathbb{R}^{n,n} = V \oplus^\perp W.$$

In an orthogonal direct sum, the second summand is the orthogonal complement of the first. Hence

$$\boxed{W = V^\perp.}$$

Method 4: Matrix-Unit Orthogonal Basis Theorem

(a) Let E_{ij} denote the standard matrix units. Under the Frobenius inner product,

$$\langle E_{ij}, E_{kl} \rangle_F = \delta_{ik} \delta_{jl}.$$

Thus the Frobenius product is the ordinary Euclidean dot product after identifying a matrix with its list of entries. Left multiplication by M applies the same orthogonal transformation to each column, because $M^T M = I_n$. Hence it preserves the Euclidean dot product on each column and therefore preserves the total Frobenius inner product:

$$\boxed{\langle MA, MB \rangle_F = \langle A, B \rangle_F.}$$

(b) The symmetric subspace V has the basis

$$\{E_{ii} : 1 \leq i \leq n\} \cup \{E_{ij} + E_{ji} : 1 \leq i < j \leq n\}.$$

The skew-symmetric subspace W has the basis

$$\{E_{ij} - E_{ji} : 1 \leq i < j \leq n\}.$$

For $i < j$ and $k < \ell$,

$$\langle E_{ij} + E_{ji}, E_{k\ell} - E_{\ell k} \rangle_F = 0,$$

because the nonzero entries cancel pairwise. Also

$$\langle E_{ii}, E_{k\ell} - E_{\ell k} \rangle_F = 0.$$

Hence every basis vector of V is orthogonal to every basis vector of W , so

$$W \subseteq V^\perp.$$

The number of basis elements of W is $n(n-1)/2$, while

$$\dim V^\perp = n^2 - \frac{n(n+1)}{2} = \frac{n(n-1)}{2}.$$

Hence

$$\boxed{W = V^\perp.}$$

Method 5: Self-Adjoint Involution Eigenspace Theorem

(a) Define the left multiplication operator

$$L_M : \mathbb{R}^{n,n} \rightarrow \mathbb{R}^{n,n}, \quad L_M(A) = MA.$$

With respect to the Frobenius inner product, the adjoint of L_M is $L_{M^\top}$, because

$$\langle L_M(A), B \rangle_F = \text{trace}((MA)^\top B) = \text{trace}(A^\top M^\top B) = \langle A, L_{M^\top}(B) \rangle_F.$$

Hence

$$L_M^* L_M = L_{M^\top} L_M = L_{M^\top M} = L_I = I.$$

Therefore L_M is an orthogonal operator on the Frobenius inner product space. Thus

$$\boxed{\langle MA, MB \rangle_F = \langle A, B \rangle_F.}$$

(b) Define

$$\tau : \mathbb{R}^{n,n} \rightarrow \mathbb{R}^{n,n}, \quad \tau(A) = A^\top.$$

Then

$$\tau^2 = I,$$

so τ is an involution. Also τ is self-adjoint with respect to the Frobenius inner product, since

$$\langle \tau(A), B \rangle_F = \text{trace}(AB)$$

and

$$\langle A, \tau(B) \rangle_F = \text{trace}(A^\top B^\top) = \text{trace}((BA)^\top) = \text{trace}(BA) = \text{trace}(AB).$$

Hence

$$\langle \tau(A), B \rangle_F = \langle A, \tau(B) \rangle_F.$$

The subspace V is the +1-eigenspace of τ , and W is the -1-eigenspace of τ :

$$V = E_1(\tau), \quad W = E_{-1}(\tau).$$

For a self-adjoint operator, eigenspaces belonging to distinct eigenvalues are orthogonal. Therefore

$$V \perp W.$$

Since $\tau^2 = I$, every matrix decomposes as

$$A = \frac{A + A^\top}{2} + \frac{A - A^\top}{2},$$

where the two terms lie in V and W respectively. Hence

$$\mathbb{R}^{n,n} = V \oplus W.$$

Combining orthogonality and direct-sum decomposition gives

$$\boxed{W = V^\perp.}$$

Marking Scheme

- Part (a), 8 marks: expands $\langle MA, MB \rangle_F$ correctly and uses $M^T M = I_n$ to reduce to $\langle A, B \rangle_F$.
- Part (b), 12 marks: shows skew-symmetric matrices are orthogonal to symmetric matrices and proves equality, either by dimension counting or by orthogonal direct-sum decomposition.

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